

Data Manipulation Language (DML) Commands

Q. Discuss Data Manipulation Language (DML) commands in SQL. Write the syntax and examples for SELECT, INSERT, UPDATE, and DELETE.

[DEF] Definition to Remember

Data Manipulation Language (DML) is the subset of SQL used to insert, retrieve, modify, and delete the actual data (records) residing within database tables. Unlike DDL, DML commands are generally not auto-committed -- changes can be rolled back until a COMMIT is issued. (CRUD: Create = INSERT, Read = SELECT, Update = UPDATE, Delete = DELETE)

DML Commands

1. INSERT

Adds new

2. SELECT (Data Query Language -- DQL)

Retrieves

3. UPDATE

Modifies

[NOTE] Warning: Omitting WHERE updates ALL rows in the table!

4. DELETE

Removes

[NOTE] Warning: Omitting WHERE deletes ALL rows (table structure remains -- unlike DROP).

DML vs DDL Comparison

Feature	DML	DDL
Operates On	Data (rows/records)	Structure (tables/schemas)
Auto-Committed	Usually **No** (can ROLLBACK)	**Yes** (permanent)
Commands	SELECT, INSERT, UPDATE, DELETE	CREATE, ALTER, DROP, TRUNCATE
Reversible	Yes, with ROLLBACK	No

[MUST-WRITE] Must-Write Points (for 10 marks)

1. DML manipulates actual data (rows/records) inside tables.
2. DML commands are NOT auto-committed -- changes can be rolled back.
3. INSERT: adds new rows; can specify column names (recommended) or use all-column syntax.
4. SELECT: retrieves data; use WHERE to filter, * for all columns.
5. UPDATE: modifies existing rows; MUST use WHERE or all rows will be updated.

6. DELETE: removes rows; MUST use WHERE or all rows will be deleted (structure preserved).

7. CRUD = Create (INSERT), Read (SELECT), Update (UPDATE), Delete (DELETE).

[RECALL] Quick Recall

DML -> Not Auto-committed -> INSERT (add rows) -> SELECT (read data) -> UPDATE (modify rows, use WHERE!) -> DELETE (remove rows, use WHERE!) -> CRUD